



# **PLANTATION JOURNAL**



## **“SUSTAINABLE FOREST & LIVELIHOOD (SUFAL) PROJECT”**

**MANGROVE PLANTATION F.Y. (2019-20)**

**PLANTATION AREA = 300.0 HEC.**

**NUMBER OF SEEDLINGS = 13,33,200 NOS.**

**CHAR KALAM BEAT, JAHAJMARA RANGE  
COASTAL FOREST DIVISION, NOAKHALI.**

## PLANTATION JOURNAL

### “SUSTAINABLE FORESTS & LIVELIHOODS (SUFAL) PROJECT”

| A Basic Information                     |                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                  |
|-----------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.                                      | Name of the plantation with Year                                                           | Mangrove Plantation (2019-2020)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                  |
| 2.                                      | Plantation Area                                                                            | 300.0 Hectare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                  |
| 3.                                      | Range                                                                                      | Jahajmara Range, Coastal Forest Division Noakhali                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                  |
| 4.                                      | Species                                                                                    | <u>Sonneratia apetala</u> (Keora), <u>Excoecaria agallocha</u> (Gewa)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                  |
| 5.                                      | No. of seedlings                                                                           | 13,33,200 Nos                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                  |
| 6.                                      | Location                                                                                   | District: Noakhali                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Union: N/A       |
|                                         |                                                                                            | Upazila: Hatiya                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Char: Char Kalam |
| 8.                                      | Name & Designation of the Officers & stuffs Involved in the Plantation Raising Activities: | 1. Mr. Bipul Krishna Das, DCF.<br>2. Mr. Md. Firoj Alom Chowdhury, ACF (A.C)<br>3. Mr. Md. Firoj Alom Chowdhury, Forest Ranger<br>4. Mr. Md. Ferdous Khan, Forester<br>5. Himangsho Mohan Sarker, Fg<br>6. Tutul Chakma, Fg<br>7. Mr. Iashin Khandaker, Fg<br>8. Mr. Md. Shala Uddin, BM                                                                                                                                                                                                                                                                           |                  |
| B Characteristic of the Plantation Site |                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                  |
| 1.                                      | Area of the Plantation                                                                     | 300 ha Mangrove Plantation in the year of 2019-2020.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                  |
| 2                                       | Edaphic Condition                                                                          | Physiographically the Char is part of the young Meghna Estuarine Flood Plain. Hobgoblin, the site is soft and clayey and it is also comparatively compact and harder as well as at the same time subjected to more saline condition. These soils permit easy penetration of mound and mangrove tree roots because of being quite fertile.                                                                                                                                                                                                                          |                  |
| 3                                       | Climate Condition                                                                          | This char has a tropical monsoon climate, characterized by basically four seasons per year. The average temperature usually ranges between 11 ° to 29 ° C in winter and 21 ° to 38 ° C during summer. The average annual rainfall is 3257 mm (Islam 2009) and varies from 2500 mm to 4500 mm. Cyclones pass regularly, particularly at the beginning and at the end of the monsoon.                                                                                                                                                                                |                  |
| 4                                       | Selection Criteria of the Planting Site                                                    | The plantation site has been selected through the following criteria.<br>a. Land suitability: The land of the planting site (Char Kalam) is soft and clayey. In addition, the site is comparatively compact and harder as well as at the same time subject to more saline condition where Mangrove seedlings e.g. Keora ( <i>Sonneratia apetala</i> ) and <i>Excoecaria agallocha</i> (Gewa)are suitable for plantation.<br>b. Grazing Problem: It is known to that there entire are a number of chars in Noakhali and accreting new lands. These chars are threat |                  |

|   |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |                   | <p>to the grazing problems. But this is free from such unavoidable phenomenon. That is why this site is selected for the mangrove plantation.</p> <p>c. Biodiversity Conservation: The Coastal afforestation program will bring a result in the establishment of plantation rich in biodiversity and higher density of wildlife habitats. Natural regeneration of forest tree species with their associates will be enhanced. In addition, through the project area the condition of coastal forest in Bangladesh will extent.</p> <p>d. Poverty Reduction: The project activities are comprised of civil works, nursery and plantation raising and maintenance, gardening etc.; which are labor intensive and technical works. Male and female can be deployed or engaged in all these works based on their working capability; which will create employment opportunities for them and help them in alleviating poverty and fulfilling needs. Thus they will be benefited socio-economically and their social esteem will be elevated.</p> <p>e. Sectoral Target: The people residing in the project area are exposed to frequent cyclones and tidal surge which makes them highly vulnerable to recent climate change effects. As a result, their socio-economic condition is less secured. Beside this, the land mass developed due to accretion of the sediments discharged through Meghna estuary needs to be stabilized. On the other hand, the objectives of the forestry sector is to increase the total forest cover of the country and to optimize sustainable production of goods and services from the existing limited forest resources. The coastal area of the proposed project is such type of land where forestry resources could be developed through establishment of plantations.</p> <p>f. National Strategy: The proposed project is also aligned with the National Strategy; the <i>Bangladesh Sixth Five Year Plan (FY2019-FY2023)</i>. According to the plan, existing scattered coastal areas will be planted to increase productivity by promoting scientific management principles. In order to prevent the extent of damage by cyclones and tidal surges, the plan envisages creating a Coastal Green Belt and enhancing access to input materials, including seeds and seedlings. In addition, past climate disasters have shown that resilience to cyclones and storm surges is greater where mangrove greenbelts and buffer ecosystems exist. It is evident that mangroves can mitigate or reduce risk of natural disasters such as cyclones and tsunamis.</p> <p>g. Global Aspect: According to the IPCC report of 2013, approximately 95% human beings are the responsible for the climate change and Bangladesh is the home to the world's largest delta systems. Two third of the country is less than five meters above the sea level. It is vulnerable to short term climate shocks such as cyclone (DFID-2011). As a result, human beings and natural resources are threats to climate change. So, there is no alternative without afforestation and reforestation in the coastal zones. That is why <b>"Char Kalam"</b> is selected for mangrove plantation.</p> |
| 5 | Climate Resilient | ★ The Coastal forest, breakwater, seawall is to absorb the impact                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

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|------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | <b>Aspect of the Plantation</b>                                      | <p>forces and to retard the flow of large storm waves and cyclone. A seawall, if tall enough, reflects the waves back out to sea. On the other hand, permeable structures, like breakwaters and coastal forests, partly reflect and partly transmit the water. In the case of a coastal forest, energy is progressively absorbed as it passes through the forest. Without the forest barrier, the tsunami/cyclone will run-up to a maximum height determined by the magnitude and nature of the seismic event that created the tsunami and local factors such as the coastal profile, offshore bathymetry and beach slope that modify the wave's force. Once the tsunami comes on shore, the amount of reduction in water depth, velocity, and force depends on how much water is reflected and energy is absorbed by the coastal forest.</p> <p>★ Through the afforestation in the project area biodiversity conservation, human security and so on will be ensured in the project area. Therefore, greenbelt of the project area by rising said plantation has been planned to bring back the balance of eco-system into the same char lands and to develop a friendly environment by raising coastal afforestation. It is for the fact that the land is suitable for mangrove and non-mangrove plantations where bio-physical and socio-economic conditions will improve. The New Char Kalam will be brought to work as a barrier during natural disasters by dint of the mangrove plantation.</p> |
| <b>C Planting Activities</b> |                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 1                            | <b>Delineation of the Planting Area</b>                              | <p>The New Char Kalam is the ultimate result of sedimentation lead to form a marshy land whose soils permit easy penetration of mangrove tree roots and agricultural crop roots because of being quite fertile. The proposed project will be used for different types of plantations which are suitable in the coastal zone by leveling with the existing species diversity. If the coastal forest lands of a target area are properly replenished with greens, the existing wildlife habitats will increase and will be a nice environment for Biodiversity conservation. This type of coastal forest lands can fight back the prevailing environmental threats and mitigate the negative impacts of the climate changes took place in these coastal zones. Thus, this type of coastal forest lands not only save the life and resources from natural disasters but also play important role in economic development.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 2                            | <b>GPS Reading of all Corners of the Planting Site</b>               | <p>The GPS has been experiment in reaching the every position of the plantation site particularly through the each corner of the site in presence of the RO, BO, FG, BM as well as all other FD-Noakhali under Hatiya Upazila.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 3                            | <b>Post-Planting Survey, Surveyor, Survey Technique &amp; Record</b> | <p>Mr. Md. Shafiqul Islam<br/>Surveyor</p> <p>The Survey has been completed through the technique of GPS reading and it is 1<sup>st</sup> recorded in the note book after that the record is shown in the Map</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

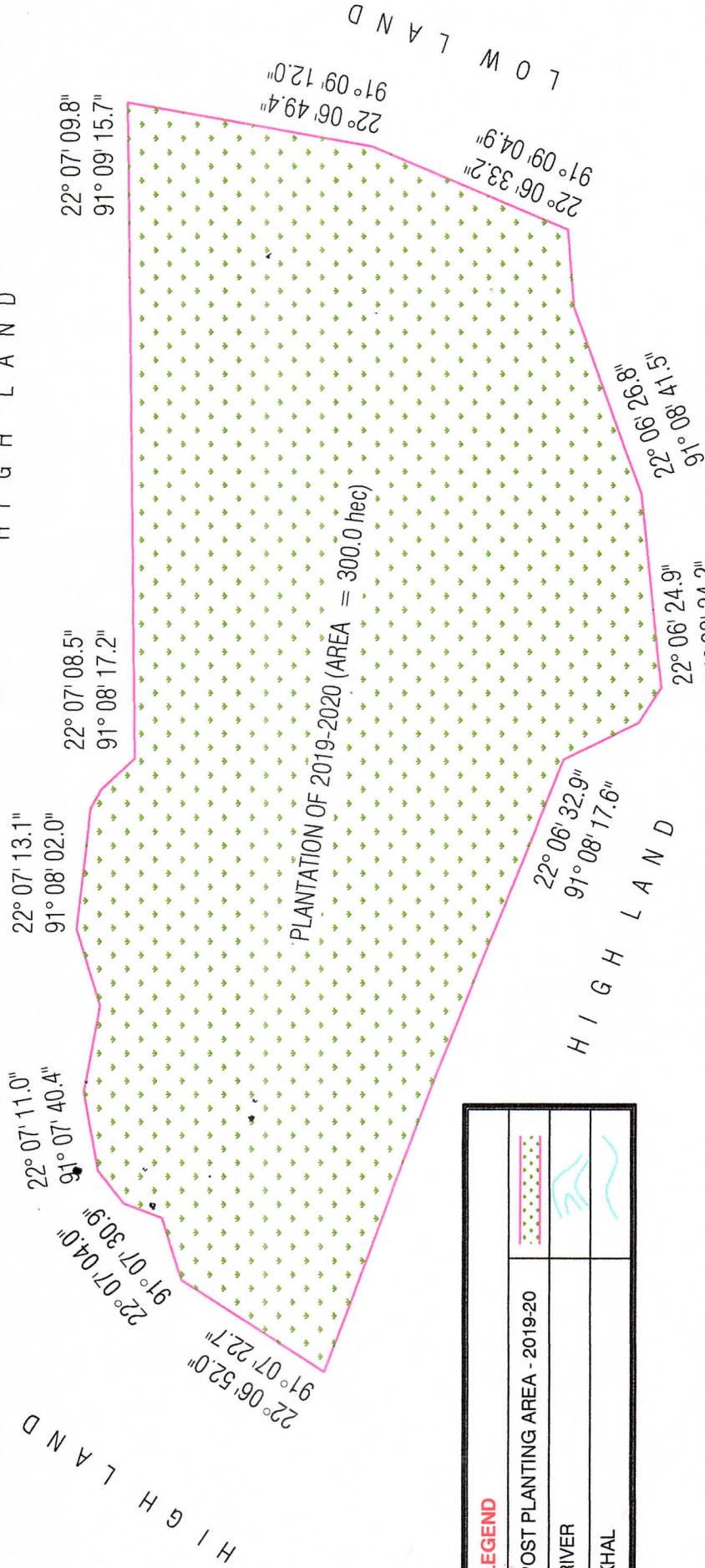
|   |                                  |                                                     |
|---|----------------------------------|-----------------------------------------------------|
| 4 | <b>Post-Planting Map (Scale)</b> | Post-Planting Map has been Attached in this Journal |
|---|----------------------------------|-----------------------------------------------------|

|    |                                                                                |                                                                                                                                                                                                                     |                                                                                                                                                                                                                     |
|----|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5. | <b>Signature of the officers Involved in the Raising Plantation Activities</b> | <p>Beat Officer, Sadar Beat,<br/>Jahajmara Range<br/>Coastal Forest Division, Noakhali.</p>                                                                                                                         | <br><b>Md. Ferdous Khan</b><br>Forester<br>Beat Officer, Char Kalam Beat<br>Jahajmara Range<br>Coastal Forest Division, Noakhali |
|    |                                                                                | <br>Range Officer<br>Jahajmara Range<br>Coastal Forest Division, Noakhali.                                                         | <br><b>Md. Firoj Alom Chowdhury</b><br>Range Officer<br>Jahajmara Range<br>Coastal Forest Division, Noakhali                     |
|    |                                                                                | <br>Mr. Md. Firoj Alom Chowdhury<br>Forest Ranger<br>Assistant Conservator of Forest (A.C).<br>Coastal Forest Division, Noakhali. | <br><b>Md. Firoj Alom Chowdhury</b><br>Assistant Conservator of Forest (A.C)<br>Coastal Forest Division, Noakhali.             |
|    |                                                                                | <p>Mr. Bipul Krishna Das<br/>DCF.<br/>Divisional Forest Officer<br/>Coastal Forest Division, Noakhali</p>                                                                                                           | <br><b>Bipul Krishna Das</b><br>Divisional Forest Officer<br>Coastal Forest Division, Noakhali.                                |

**POST PLANTING MAP OF MANGROVE PLANTATION AREA**  
**BEAT : CHAR KALAM BEAT , RANGE : JAHAJMARA RANGE**  
**DIVISION : COASTAL FOREST DIVISION, NOAKHALI.**  
**NAME OF CHAR: CHAR KALAM, JAHAJMARA, HATIYA, NOAKHALI.**

**YEAR : 2019 - 2020**  
**AREA = 300.0 HEC**  
**SCALE = 1:16000**

**PROJECT: SUSTAINABLE FORESTS & LIVELIHOODS (SUFAL) PROJECT**



| LEGEND                       |  |
|------------------------------|--|
| POST PLANTING AREA - 2019-20 |  |
| RIVER                        |  |
| KHAL                         |  |

Approved by

Inspected by

Counter Sign

Submitted by

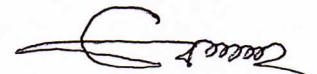
মোঃ ফেরাউজ খান  
কর্ত্তার  
বিট কর্মকর্তা, চর কালাম বিট  
জাহাজমারা রেঞ্জ।

২০১৯-২০২০ আর্থিক সনে ৩০০.০ হেক্টর ম্যানগ্রোভ বাগান সৃজন।

কন্ট্রোল নাম : টেকসই বন ও জীবিকা (সুফল) প্রকল্প :

| S. No.                   | Voucher No & Month       | Amount     |
|--------------------------|--------------------------|------------|
| 1                        | 1/Jama of January/2020   | 262500.00  |
| 2                        | 2/Jama of January/2020   | 262500.00  |
| 3                        | 3/Jama of January/2020   | 262500.00  |
| 4                        | 4/Jama of January/2020   | 24000.00   |
| 5                        | 5/Jama of January/2020   | 24000.00   |
| 6                        | 6/Jama of January/2020   | 24000.00   |
| 7                        | 7/Jama of January/2020   | 24000.00   |
| 8                        | 8/Jama of January/2020   | 24000.00   |
| 9                        | 9/Jama of January/2020   | 24000.00   |
| 10                       | 10/Jama of January/2020  | 24000.00   |
| 11                       | 11/Jama of January/2020  | 24000.00   |
| 12                       | 12/Jama of January/2020  | 24000.00   |
| 13                       | 13/Jama of January/2020  | 24000.00   |
| 14                       | 14/Jama of January/2020  | 24000.00   |
| 15                       | 15/Jama of January/2020  | 24000.00   |
| 16                       | 16/Jama of January/2020  | 24000.00   |
| 17                       | 17/Jama of January/2020  | 18000.00   |
| 18                       | 1/Jama of February/2020  | 225000.00  |
| 19                       | 2/Jama of February/2020  | 244000.00  |
| 20                       | 3/Jama of February/2020  | 243500.00  |
| 21                       | 4/Jama of February/2020  | 24000.00   |
| 22                       | 5/Jama of February/2020  | 24000.00   |
| 23                       | 6/Jama of February/2020  | 24000.00   |
| 24                       | 7/Jama of February/2020  | 24000.00   |
| 25                       | 8/Jama of February/2020  | 24000.00   |
| 26                       | 9/Jama of February/2020  | 24000.00   |
| 27                       | 10/Jama of February/2020 | 24000.00   |
| 28                       | 11/Jama of February/2020 | 24000.00   |
| 29                       | 12/Jama of February/2020 | 24000.00   |
| 30                       | 13/Jama of February/2020 | 24000.00   |
| 31                       | 14/Jama of February/2020 | 24000.00   |
| 32                       | 15/Jama of February/2020 | 24000.00   |
| 33                       | 16/Jama of February/2020 | 24000.00   |
| 34                       | 17/Jama of February/2020 | 18000.00   |
| Sub-Total =              |                          | 2160000.00 |
| Contractor (Materials) = |                          | 105000.00  |
| Grant Total =            |                          | 2265000.00 |

  
Md. Firoz Alam Chowdhury  
Forest Ranger  
Range Officer, Comptroller Range  
Jahajmara, Netrokona

  
মোঃ ফেরদাউস খান  
ফরেস্টার  
বিট কর্মকর্তা, চর কালাম বিট  
জাহাজমারা রেঞ্জ।